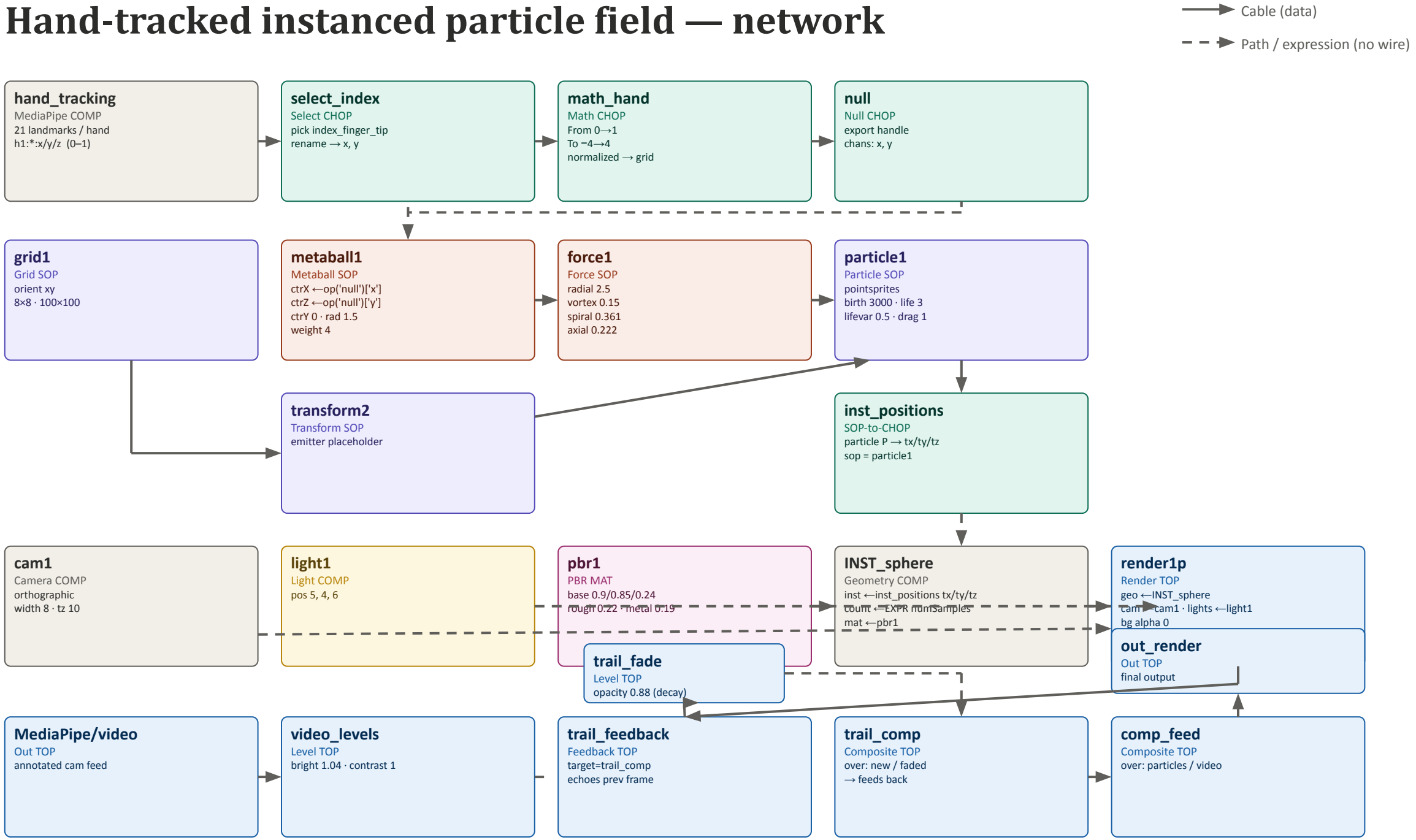
WEEK 4 OF 4

Putting It Together

Your hands, your particles, and making it look like something

Both halves already exist. Today is one connection, then an hour of craft.

Hand-tracked instanced particle field — network



What each node does

hand_tracking *MediaPipe COMP*
Runs hand detection on the webcam, outputs 21 landmarks per hand as normalized 0–1 channels.

select_index *Select CHOP*
Picks just the index-fingertip channels and renames them to clean x, y.

math_hand *Math CHOP*
Remaps 0–1 screen space to −4…4 world units so the finger reaches the whole grid.

null *Null CHOP*
Stable handle the metaball references — lets you rebuild upstream without breaking the link.

grid1 *Grid SOP*
The point field, upright in the xy plane. 100×100 = 10,000 source points.

metaball1 *Metaball SOP*
An invisible force-field shape. Its Center follows the fingertip; its radius is the reach of the force.

force1 *Force SOP*
Reads the metaball field and turns it into radial + vortex + spiral forces on the particles.

particle1 *Particle SOP*
Simulates the particles: born on the grid, pushed by the force, expiring after their life.

transform2 *Transform SOP*
Identity placeholder between grid and particles for repositioning the emitter later.

inst_positions *SOP-to-CHOP*
Converts particle point positions (P) into tx/ty/tz channels the instancer can read.

INST_sphere *Geometry COMP*
Stamps a sphere at every particle position via instancing. numinstances tracks the live count.

pbr1 *PBR MAT*
Light-reactive material — gives each sphere a different shade depending on how it faces the light.

light1 *Light COMP*
The light source PBR needs. Off to one side so shading varies across the field.

cam1 *Camera COMP*
Orthographic, facing the grid — maps the field 1:1 to the screen, no perspective distortion.

render1p *Render TOP*
Draws the instanced spheres with the camera and light, on a transparent background.

MediaPipe/video *Out TOP*
The annotated camera image (video + drawn hand skeleton).

video_levels *Level TOP*
Brightness / contrast control on the video before compositing.

trail_feedback *Feedback TOP*
Hands back the previous frame (target = trail_comp) so trails can accumulate.

trail_fade *Level TOP*
Dims the fed-back history each pass — this decay is what makes trails fade.

trail_comp *Composite TOP*
New particles OVER faded history; its output becomes next frame’s history (closes the loop).

comp_feed *Composite TOP*
Lays the trailed particle layer OVER the video feed.

out_render *Out TOP*
Final output of the whole network.

Building the patch, step by step

1 · Hand input

1. Drop in the MediaPipe hand-tracking component; confirm it outputs landmark channels.
2. Select CHOP → pick index_finger_tip x/y, rename to x, y.
3. Math CHOP → remap 0–1 to −4…4. Null CHOP after it as the handle.

2 · Force + particles

1. Grid SOP, orient xy, 8×8. Transform SOP after it (emitter placeholder).
2. Metaball SOP → Force SOP → into Particle SOP inputs 1 & 2.
3. Metaball Center X ← op('null')['x'], Center Z ← op('null')['y'], Center Y = 0.
4. Particle SOP: birth 3000, life 3, lifevar 0.5 — flowing field that respawns on the grid.

3 · Instancing + look

1. SOP-to-CHOP (inst_positions) from particle1 → tx/ty/tz channels.
2. Geometry COMP with a small sphere inside; Instance source = inst_positions.
3. numinstances = EXPRESSION op('inst_positions').numSamples (tracks live count!).
4. PBR MAT on the COMP + a Light COMP in the scene.

4 · Render + composite

1. Render TOP: Geometry ← INST_sphere, Camera ← cam1 (ortho), Lights ← light1, bg alpha 0.
2. Video In (MediaPipe/video) → Level TOP for brightness/contrast.
3. Feedback TOP (target = trail_comp) → Level TOP (fade) → Composite (new over faded).
4. Composite that trailed layer OVER the video → Out TOP.

Build it with me

1 Point the export at your hand

In the export table, swap the two channel names from the mouse to h1:index_finger_tip:x and :y. Wave. Something should move.

2 Fix the range

Open the Math CHOP and adjust until your hand's full reach maps to the full width of the scene. Watch the real numbers as you move.

3 Add depth

Add a third row to the export table: the z channel driving metaball1's Translate Z. Now pushing your hand toward the camera moves it through the scene.

4 Smooth it

A Lag CHOP before null_hand. Raise it until the movement feels good rather than twitchy. This is taste, not a correct answer.

5 Add the second hand

A second metaball, plus rows in the export table pointing null_tip_h2b at it, wired into the same Force SOP chain. Two hands, two pushers.

6 Save

Before you start experimenting. You will want to come back to this exact state.

Giving the particles a shape

Right now they are dots. They do not have to be.

Instancing: draw one small shape thousands of times, once per particle.

How it works

- Put a small shape inside your Geometry COMP — a Sphere, Box, or Circle SOP.
- On the geo's Instance page, turn instancing on and point it at your particles.
- Keep the shape SMALL. A sphere of radius 1 drawn ten thousand times is one solid blob.

Which shape to pick

Circle or a flat card is cheapest and looks almost identical at particle scale.

Box gives you a hard, digital feel. Sphere is softer but costs more.

Try all three. The difference is bigger than you expect.

Two warnings from Day 3, still true

Instancing makes everything heavier — if your frame rate drops, lower the grid density first. And if you swap in a lit material to get shading, you will need a Light COMP or the whole thing goes black again.

More forces

One metaball pushing is a start. Layering forces is where it gets interesting.

A second metaball

Driven by your other hand. Two pushers interacting produce far more complex motion than one.

Wind

A constant push in one direction. Makes everything drift, and gives the scene a sense of weather.

Turbulence

Random small-scale movement. Stops the motion looking mechanical. A little goes a long way.

Drag

On the Particle SOP itself. High drag makes particles heavy and syrupy; low drag makes them fly.

Add one at a time.

Stack three new forces at once and when it looks wrong you will have no idea which one did it. Add, look, keep or remove, then add the next.

Making it look good

Bloom

A Blur, then an Add composite back over the original. Bright areas glow. This single move does more for the look than anything else on this slide.

Colour

A Ramp TOP plus a Lookup, or a Level's tint controls. Particles do not have to be white.

Contrast

A Level TOP. Crush the blacks and the whole thing reads as more deliberate and less like a default render.

Trails

The feedback loop from Day 1, applied to your render. Particles leave smears behind them.

Chain them after your Render TOP, ending in a Null

render1 → blur → add → level → null_final. Same discipline as Day 1: one job per node, a named Null at the end. If you want to change the whole look later, you rebuild behind that Null and nothing else cares.

Everything that fails silently

Four weeks of hard-won list. All of these give you no error message at all.

A wrong or empty parameter path

"Address, not a wire" — if a node reference is misspelled or blank, nothing happens and nothing complains. Always check this first.

Cooking switched off

Right-click a node or component. If Cooking is unchecked it freezes on its last frame — which looks identical to still loading.

The render flag (bottom-RIGHT square)

Different from the viewer square at bottom-left. New nodes inside a Geometry COMP do not inherit it.

A Geometry COMP's default contents

It ships with a shape inside. Leave it there and it renders in front of everything you built.

Feedback collapsing your resolution

A feedback loop starts at 128x128 and drags the whole downstream chain down with it.

A lit material with no light

Renders pure black. Use a Constant MAT and the problem cannot occur.

The method that solves all six: click through your chain node by node and check what each one is actually outputting. The place where it stops making sense is the place to fix. That habit is the most transferable thing you take from this week.

Showing it to someone

Getting it out of the editor and onto a screen.

Perform Mode

- Add a Window COMP and point it at your final Null.
- Press F1, or use the Perform button, to go fullscreen with no interface.
- Escape gets you back out.
- This is what you would actually run at an exhibition.

Recording it

A Movie File Out TOP after your final Null captures video straight to disk.

For documentation, a plain screen recording is usually easier and good enough — and it captures you moving, which is the point.

Your deliverable

A file that runs in Perform Mode, responds to at least two things your hands do, and a thirty-second recording of you using it. Post both. That is the course finished.

Where to go from here

What we deliberately skipped, and what it unlocks.

Python

Automate anything repetitive, build custom interfaces, respond to events. The natural next step if you are technical.

GLSL shaders

Write your own effects instead of combining existing ones. Steep, but it is where the distinctive looks come from.

Audio reactivity

An Audio Device In CHOP is not much harder than the Mouse In you used. Sound driving particles is a whole genre.

POPs

A newer, GPU-native particle system alongside the SOP one you learned. Worth exploring once the concepts are solid.

Building components

Package your own work as a .tox, the way MediaPipe was handed to you. Reuse across projects.

Sending signals out

OSC, DMX, NDI — driving lights, other software, or a second machine. This is how installations get built.

You now know enough to learn the rest. Every item above is a variation on things you have already done: nodes, wires, addresses, and reading what each step is actually outputting.

That's a wrap!

You now have a working interactive system: a camera watching you, a computer finding your hands in the image, thousands of particles responding to where you point, and a finished image on screen. Every piece of that you built yourself, and you know why each node is there.

Post your final recording. Keep the file — it is a starting point, not an exercise.

Thank you!